

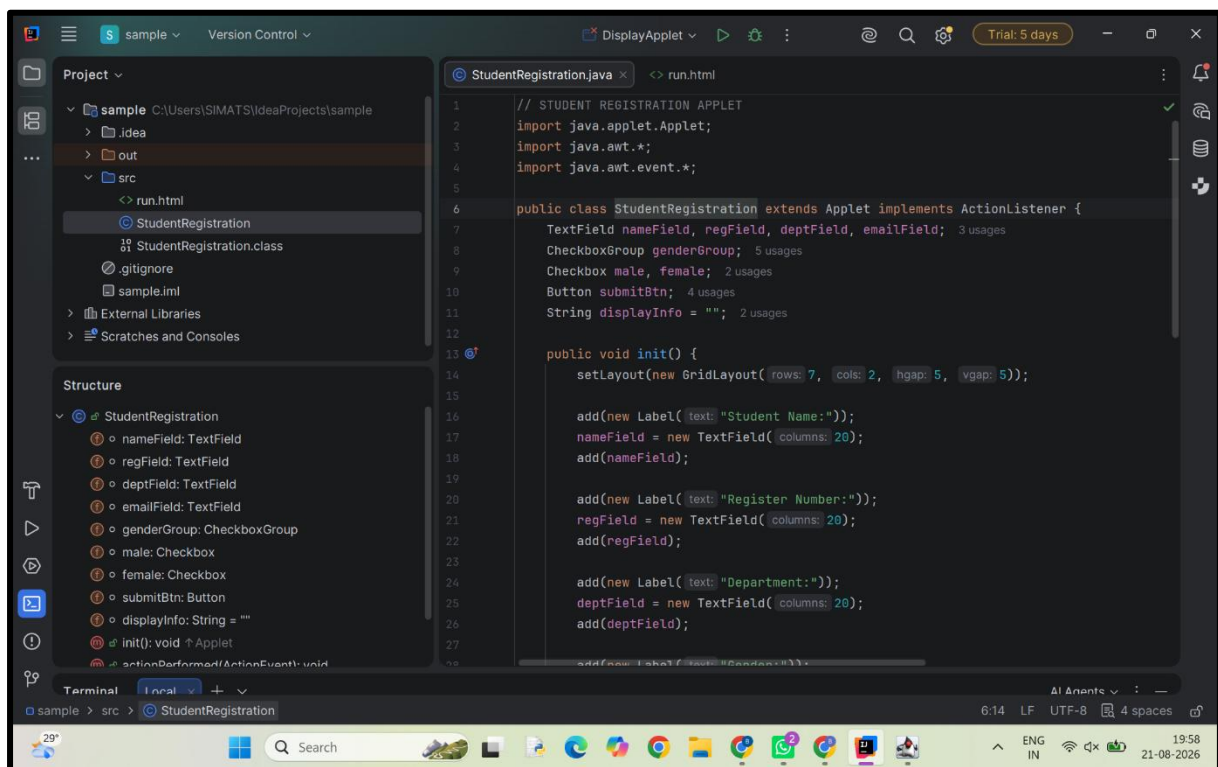
CO4 AT2 LABRORATORY PERFORMANCE

1.STUDENT REGISTRATION APPLET

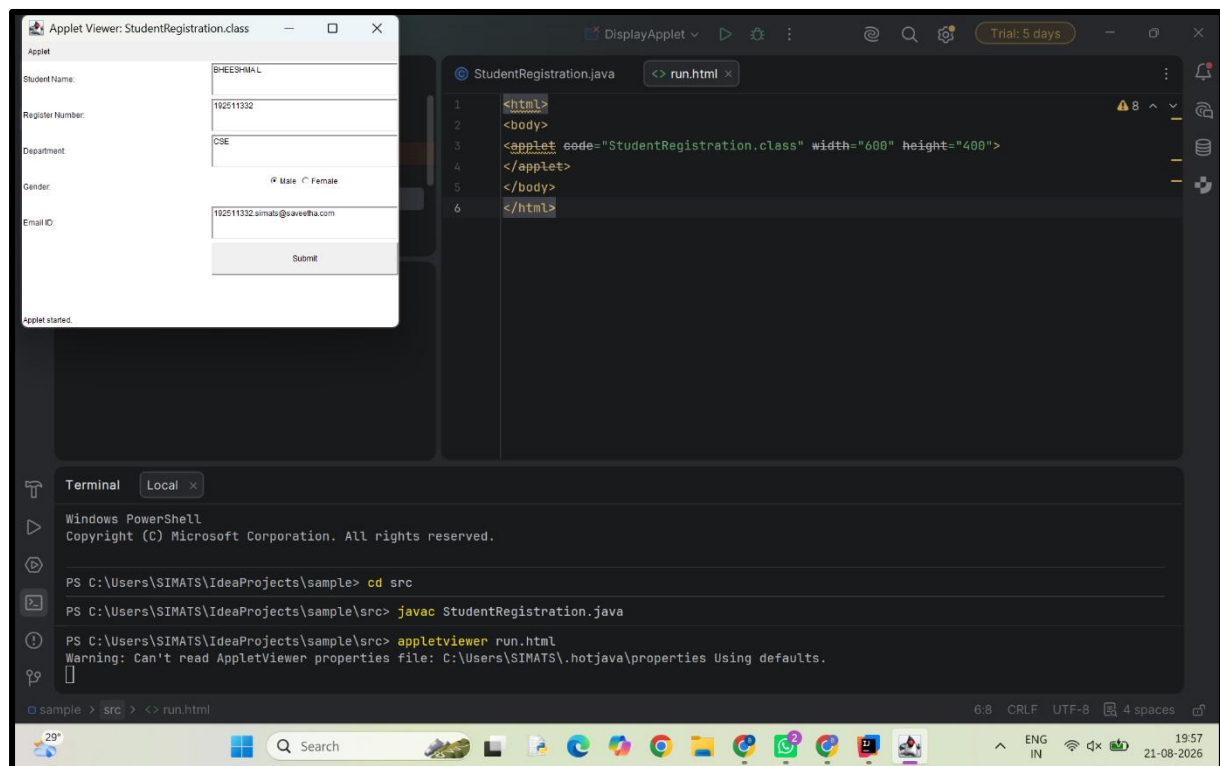
Develop a Java Applet for a university student registration form. The applet should display labels and input controls for Student Name, Register Number, Department, Gender, and Email ID. The application should use appropriate AWT controls such as Label, TextField, Checkbox, CheckboxGroup, and Button. Use a suitable layout manager to organize the controls. When the student clicks the Submit button, the entered information should be displayed within the applet window.

The student must demonstrate:

1. Applet initialization and architecture.
2. AWT controls.
3. Layout management.
4. Button event handling.
5. Appropriate event listener interface.
6. Display of user-entered information.



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2.SIMPLE CALCULATOR APPLLET

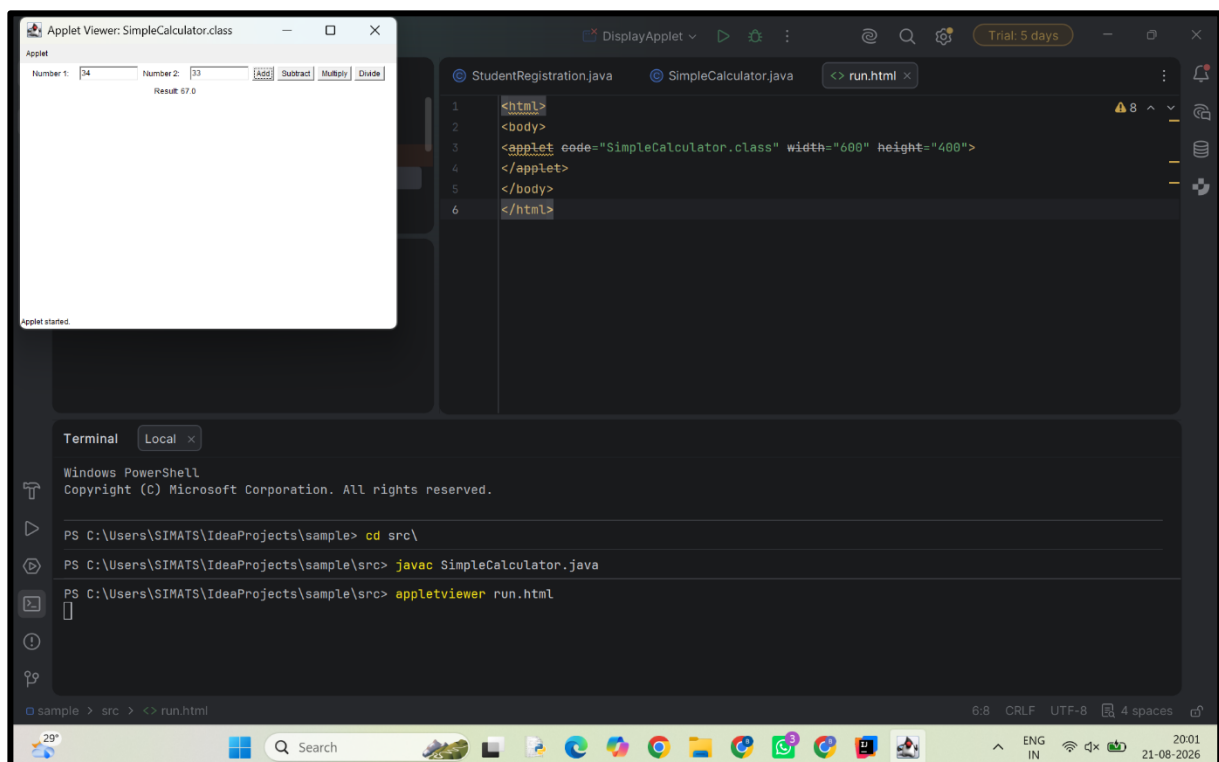
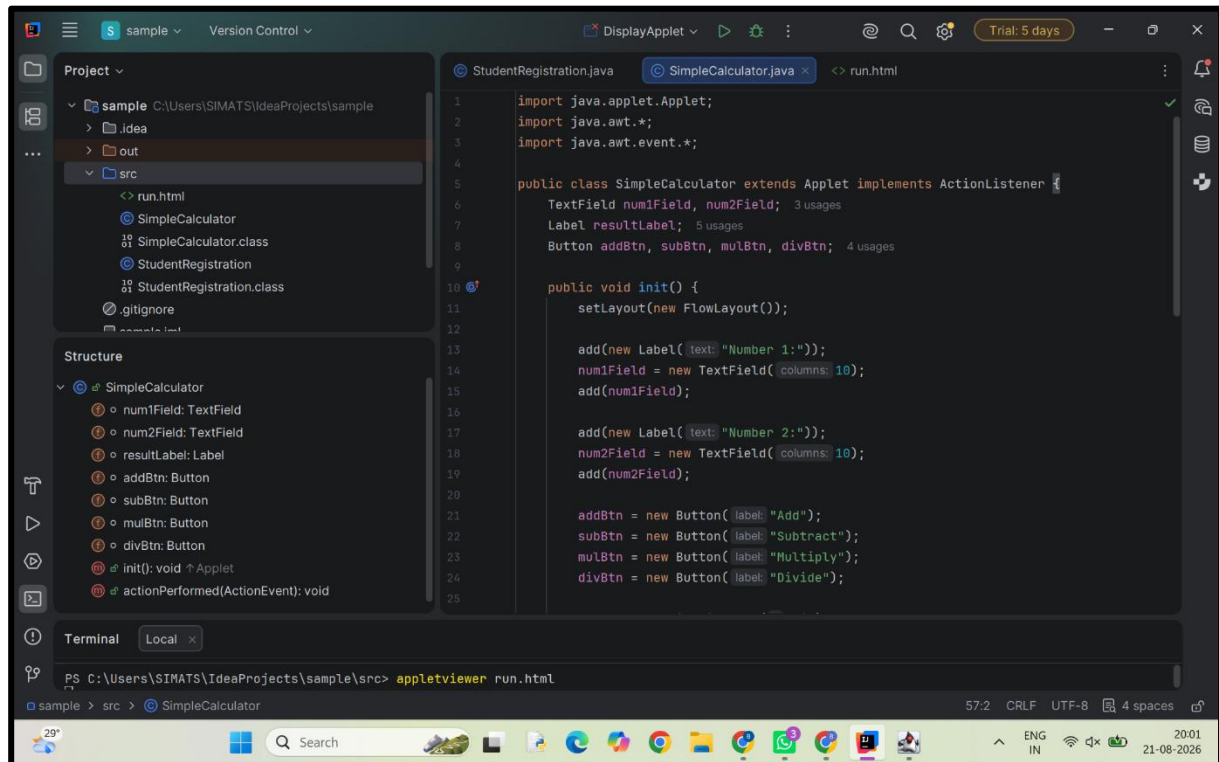
Develop an Applet-based calculator that accepts two numbers and provides buttons for Addition, Subtraction, Multiplication, and Division The application should use TextField, Label, and Button controls. Appropriate event listeners must be implemented so that clicking each button performs the corresponding operation.

The program must handle invalid numeric input and division by zero appropriately

The student must demonstrate:

1. Basic Applet structure.
2. AWT controls.
3. Event sources and event objects
4. ActionListener.
5. Exception handling.
6. Display methods.

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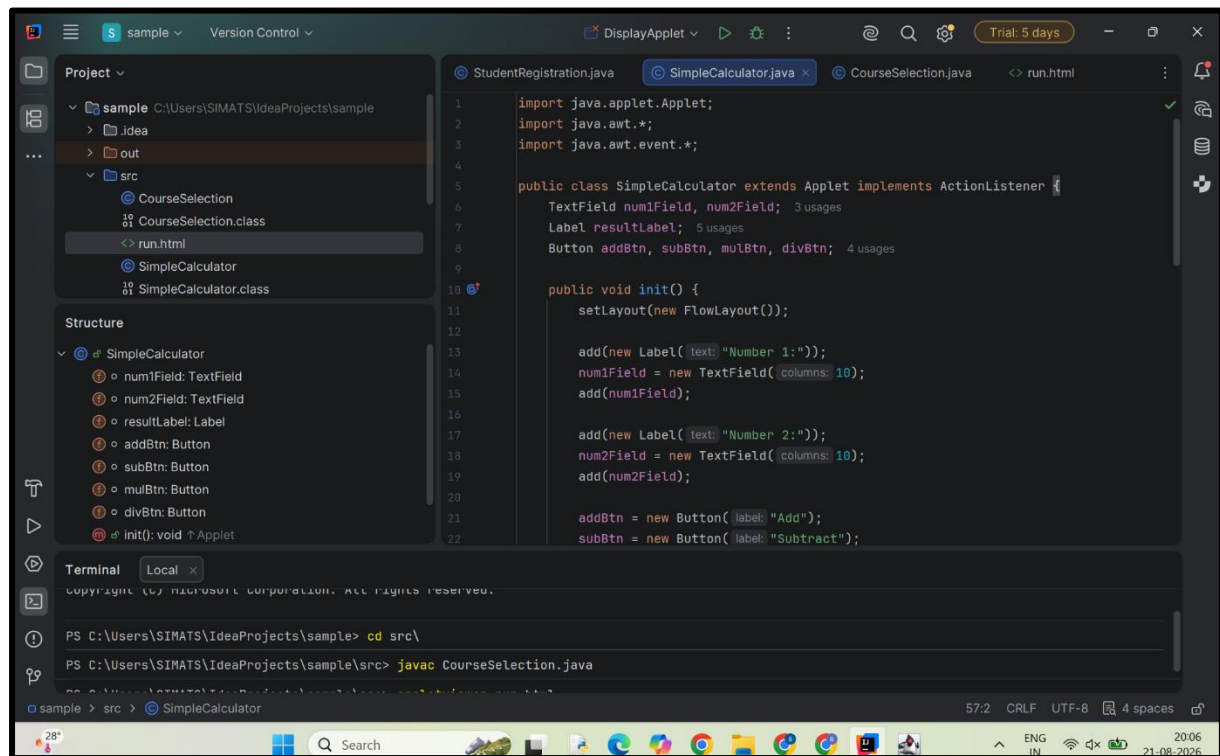
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3.COURSE SELECTION APPLET

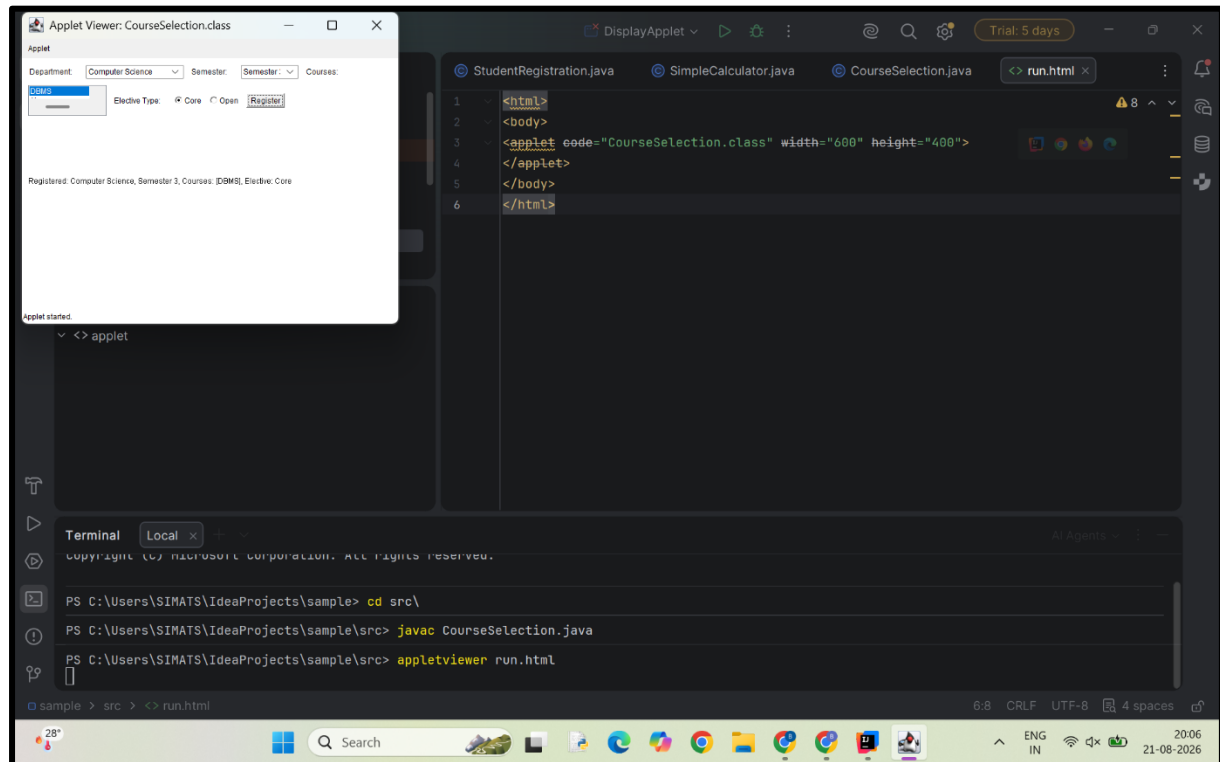
Develop a university course selection Applet where students can select:

- Department
- Semester
- Course
- Elective type

Use Choice, List, Checkbox, and Button controls wherever appropriate. When the student clicks Register, display the selected courses and other information. Use appropriate layout managers and event listeners.



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4. Traffic Signal Control Applet

Develop an AWT Applet that simulates a traffic signal with three options:

Red

Yellow

Green

Use appropriate controls to allow the user to select a signal. When the user selects a signal, display the corresponding message such as

STOP, WAIT, or GO.

Use appropriate event sources, event classes, and listener interface

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The screenshot shows an IDE with the following components:

- Project Explorer:** Lists files including CourseSelection.class, run.html, SimpleCalculator, StudentRegistration, StudentRegistration.class, TrafficSignal, TrafficSignal.class, and .gitignore.
- Structure:** Shows the TrafficSignal class structure with members: lightGroup (CheckboxGroup), red (Checkbox), yellow (Checkbox), green (Checkbox), message (String), redColor (Color), yellowColor (Color), and greenColor (Color).
- Code Editor:** Displays the TrafficSignal.java file with the following code:

```
5 public class TrafficSignal extends Applet implements ItemListener {
47 }
48
49 @
50
51 public void paint(Graphics g) {
52     // Draw traffic light box
53     g.setColor(Color.BLACK);
54     g.fillRect(x: 50, y: 50, width: 60, height: 160);
55
56     // Draw lights
57     g.setColor(redColor);
58     g.fillOval(x: 60, y: 60, width: 40, height: 40);
59
60     g.setColor(yellowColor);
61     g.fillOval(x: 60, y: 110, width: 40, height: 40);
62
63     g.setColor(greenColor);
64     g.fillOval(x: 60, y: 160, width: 40, height: 40);
65
66     // Draw message
67     g.setColor(Color.BLACK);
68     g.setFont(new Font(name: "Arial", Font.BOLD, size: 20));
69     g.drawString(str: "Signal: " + message, x: 150, y: 130);
```
- Terminal:** Shows the execution commands:

```
PS C:\Users\SIMATS\IdeaProjects\sample> cd src\
PS C:\Users\SIMATS\IdeaProjects\sample\src> javac TrafficSignal.java
PS C:\Users\SIMATS\IdeaProjects\sample\src> appletviewer run.html
```

The screenshot shows the IDE with the following components:

- Applet Viewer:** A window titled "Applet Viewer: T..." showing a traffic light with the yellow light illuminated. The text "Signal: WAIT" is displayed below the lights. The status bar indicates "Applet started."
- Code Editor:** Displays the run.html file with the following code:

```
1 <html>
2 <body>
3 <applet code="TrafficSignal.class" width="400" height="400">
4 </applet>
5 </body>
6 </html>
```
- Terminal:** Shows the execution commands:

```
PS C:\Users\SIMATS\IdeaProjects\sample> cd src\
PS C:\Users\SIMATS\IdeaProjects\sample\src> javac TrafficSignal.java
PS C:\Users\SIMATS\IdeaProjects\sample\src> appletviewer run.html
```